DATA 110 · FALL 2026

# Seven examples

**Rebin Muhammad**

Montgomery College

Palmer penguins, the eight planets, normal temperatures for Death Valley and Houston, happiness scores for 142 countries, and John Snow's map of a cholera outbreak in London in 1854.

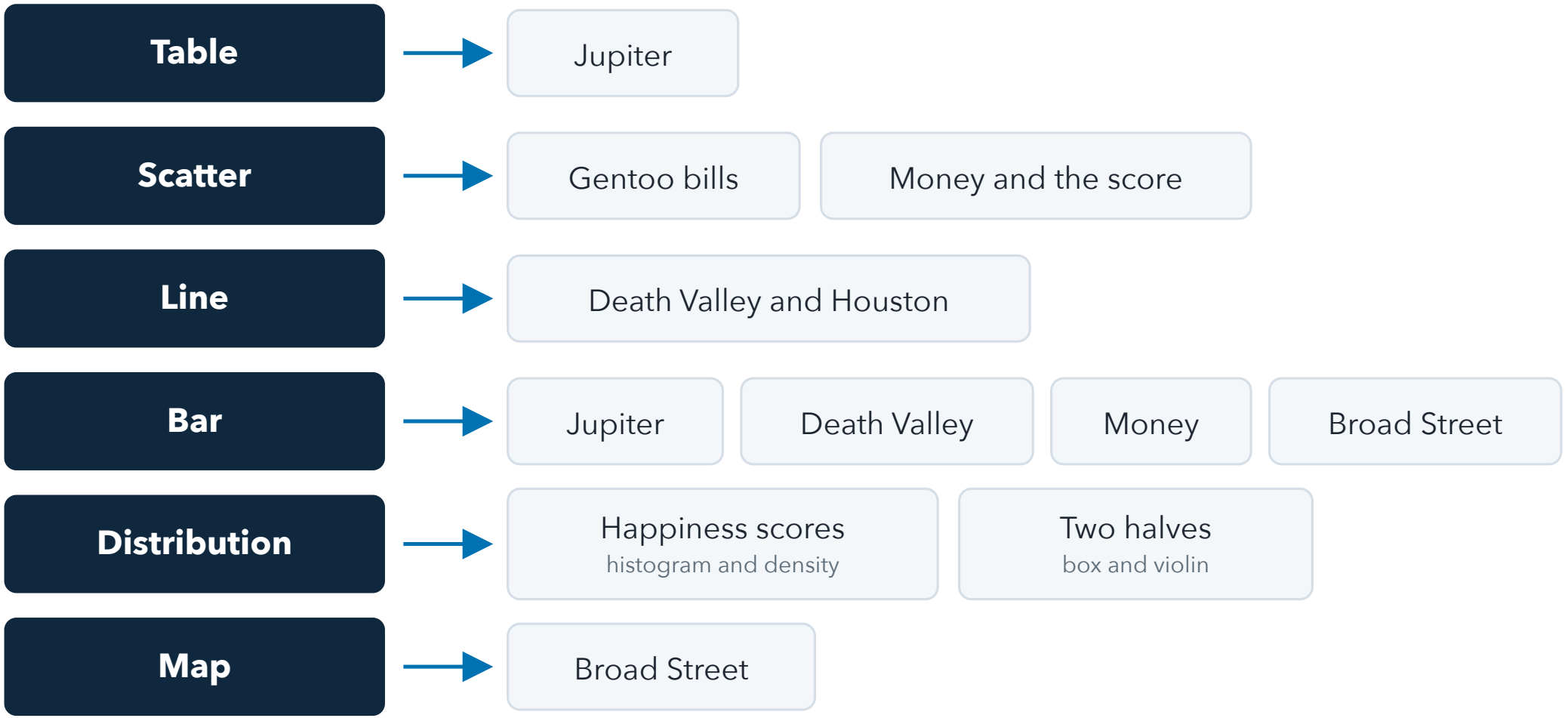
THE STORY

# Each example is a story you tell

- The question is the tension.
- The first chart shows what is going on.
- The next chart answers the piece the first chart leaves open.
- Say the result in one sentence.
- End on that sentence.

THE SEVEN

# Which chart each story uses



EXAMPLE 1

# Gentoo bills and Adelie bills

Palmer penguins. Bill length and bill depth, in millimeters.

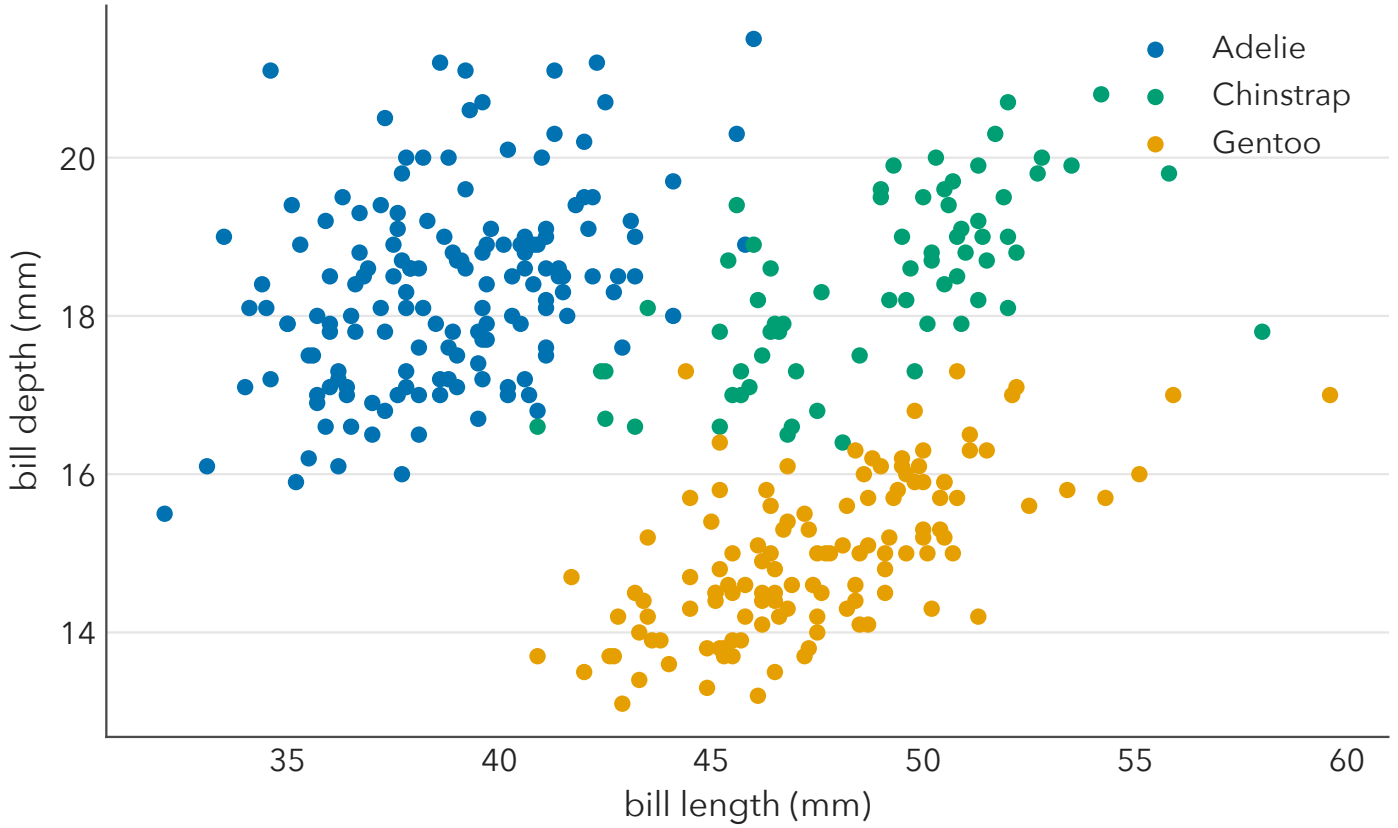
QUESTION

# Are Gentoo bills longer and shallower than Adelie bills?

Longer means a greater bill length. Shallower means a smaller bill depth.

EXAMPLE 1 · SCATTER

# Gentoo bills are longer and shallower than Adelie bills




Adelie averages 38.8 mm long and 18.3 mm deep. Gentoo averages 47.5 mm long and 15.0 mm deep.

EXAMPLE 1

# Chinstrap is outside that claim

Chinstrap bills average 48.8 mm long and 18.4 mm deep. They are long, and they are not shallow.

The chart does not say why Gentoo bills are shallow.

EXAMPLE 2

# How wide is Jupiter?

The width of each of the eight planets, measured in Earths.

QUESTION

# How many Earths, side by side, match Jupiter's width?

Earth's width is 1. The next slide lists all eight planets.

EXAMPLE 2 · TABLE

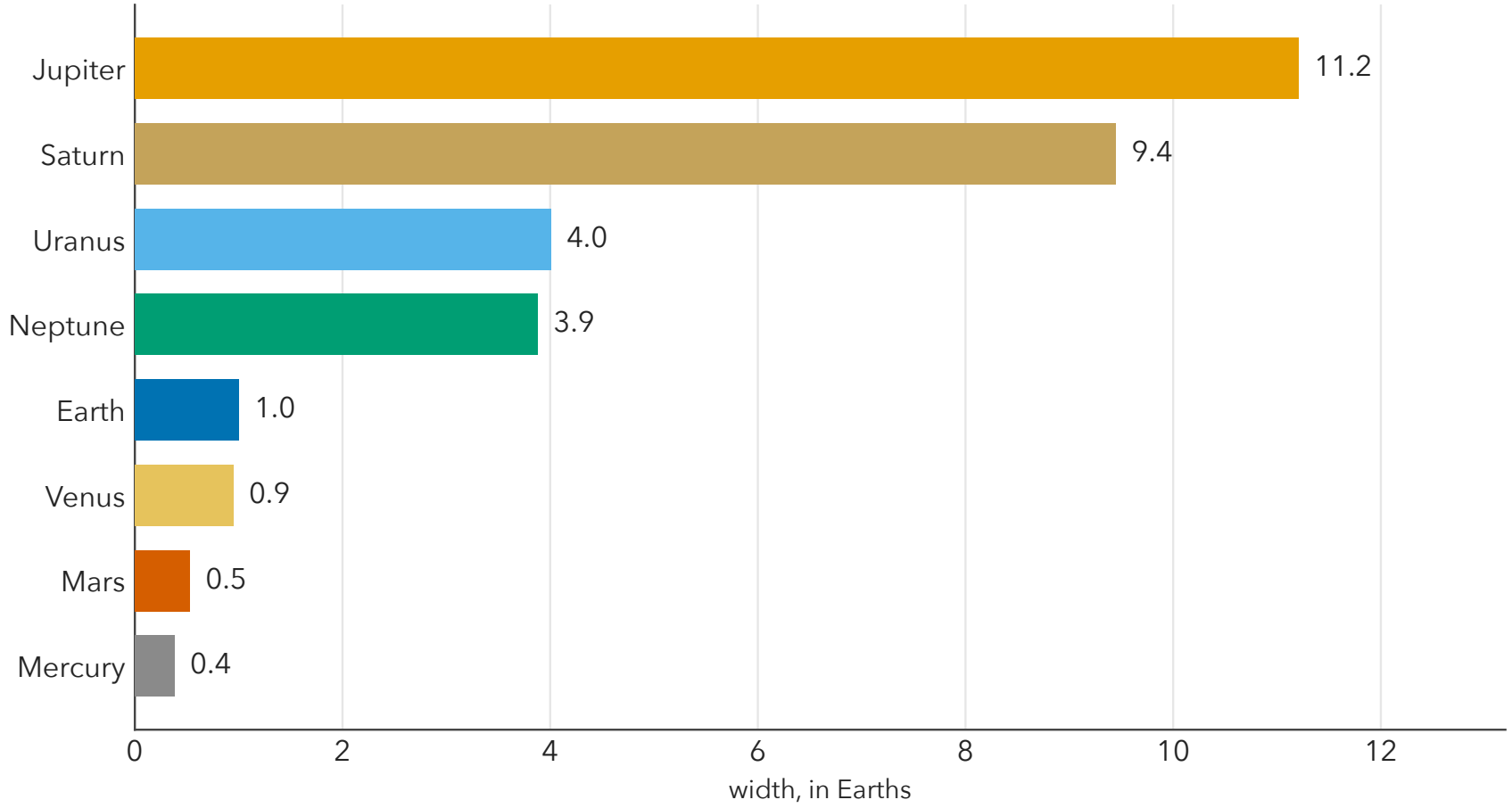
# Eight numbers, before the bar

| PLANET | RADIUS, KM | EARTHS |
|---|---|---|
| Mercury | 2,440 | 0.4 |
| Venus | 6,052 | 0.9 |
| Earth | 6,378 | 1.0 |
| Mars | 3,396 | 0.5 |
| **Jupiter** | **71,492** | **11.2** |
| Saturn | 60,268 | 9.4 |
| Uranus | 25,559 | 4.0 |
| Neptune | 24,764 | 3.9 |

Earths is the radius divided by Earth's, 6,378 km, rounded to one decimal. The gold row is Jupiter.

EXAMPLE 2 · BAR

# About 11. Jupiter is 11.2 Earths wide




Saturn is 9.4 Earths wide. Earth is 1.

EXAMPLE 2

# The rings are not in the chart

Saturn’s rings reach farther than the planet. The bar is the planet itself.

Pluto is a dwarf planet, smaller than our Moon, so it is not one of the eight.

EXAMPLE 3

# Death Valley and Houston

Normal daily temperatures, averaged into one number for each month.

QUESTION

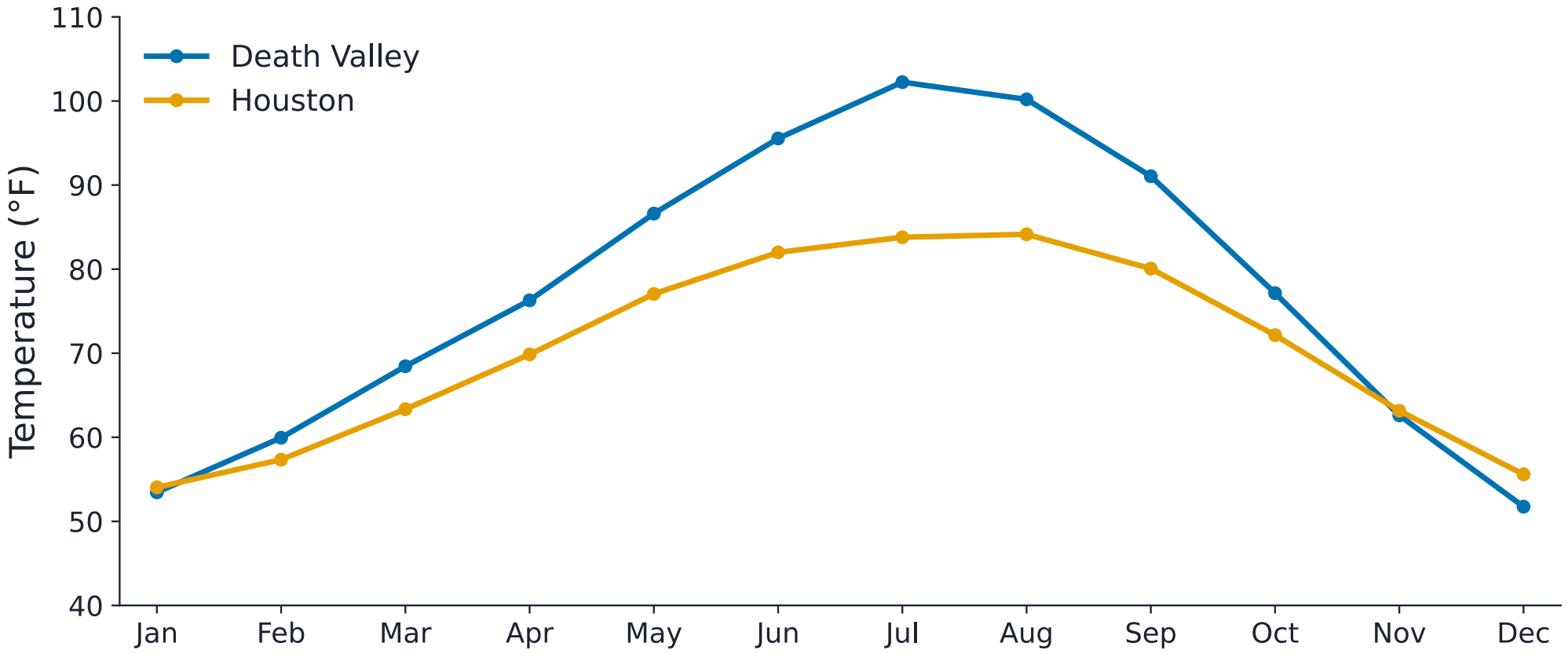
# Is Death Valley hotter than Houston every month?

The temperatures are in degrees Fahrenheit.

EXAMPLE 3 · LINE

# Death Valley is hotter for most of the year

The lines cross in the winter.

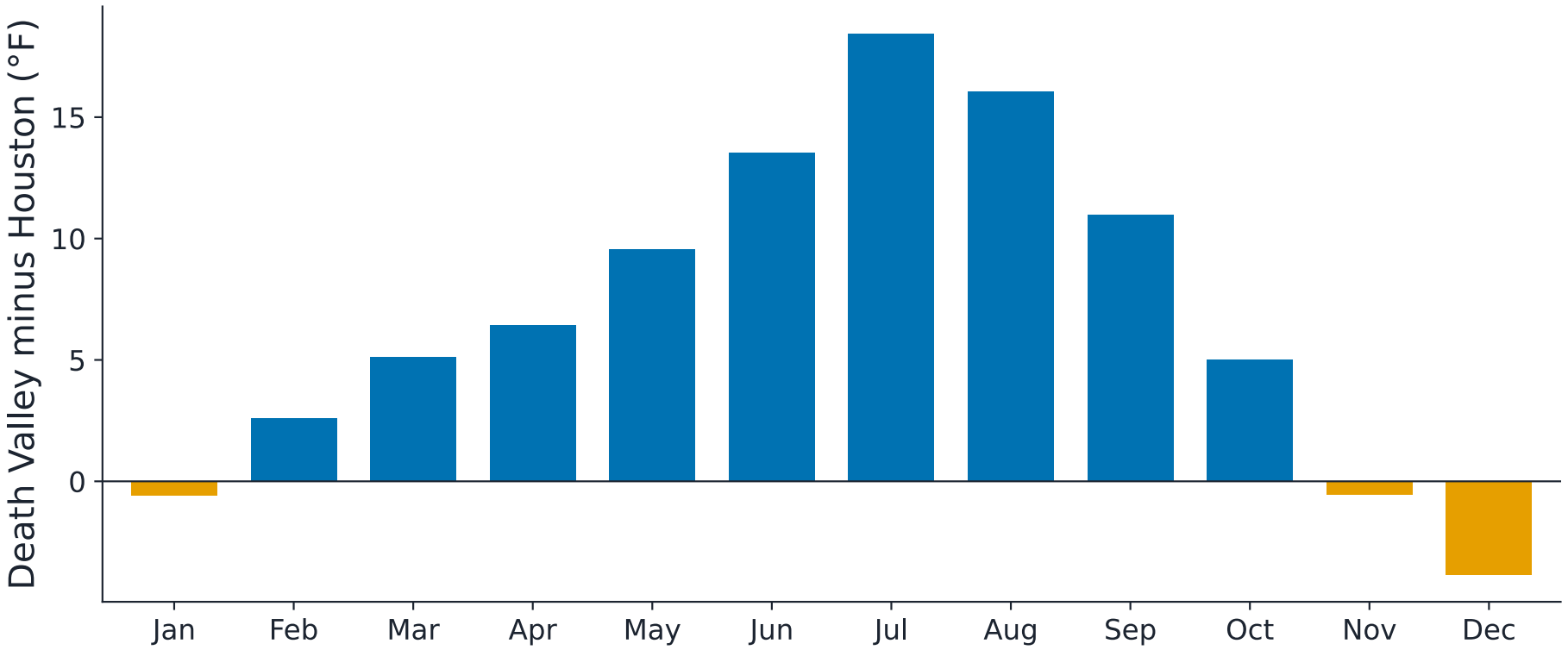



July averages 102.2°F in Death Valley and 83.8°F in Houston.

EXAMPLE 3 · BAR

# Houston is warmer in three months

Each bar is Death Valley minus Houston. A bar below zero means Houston is warmer.



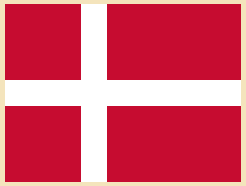

December is the widest gap the other way: Houston is 3.8°F warmer. January and November are close, and Houston is still warmer.

EXAMPLE 4

# Where do the scores sit?

Happiness scores for 142 countries. One row per country.

QUESTION



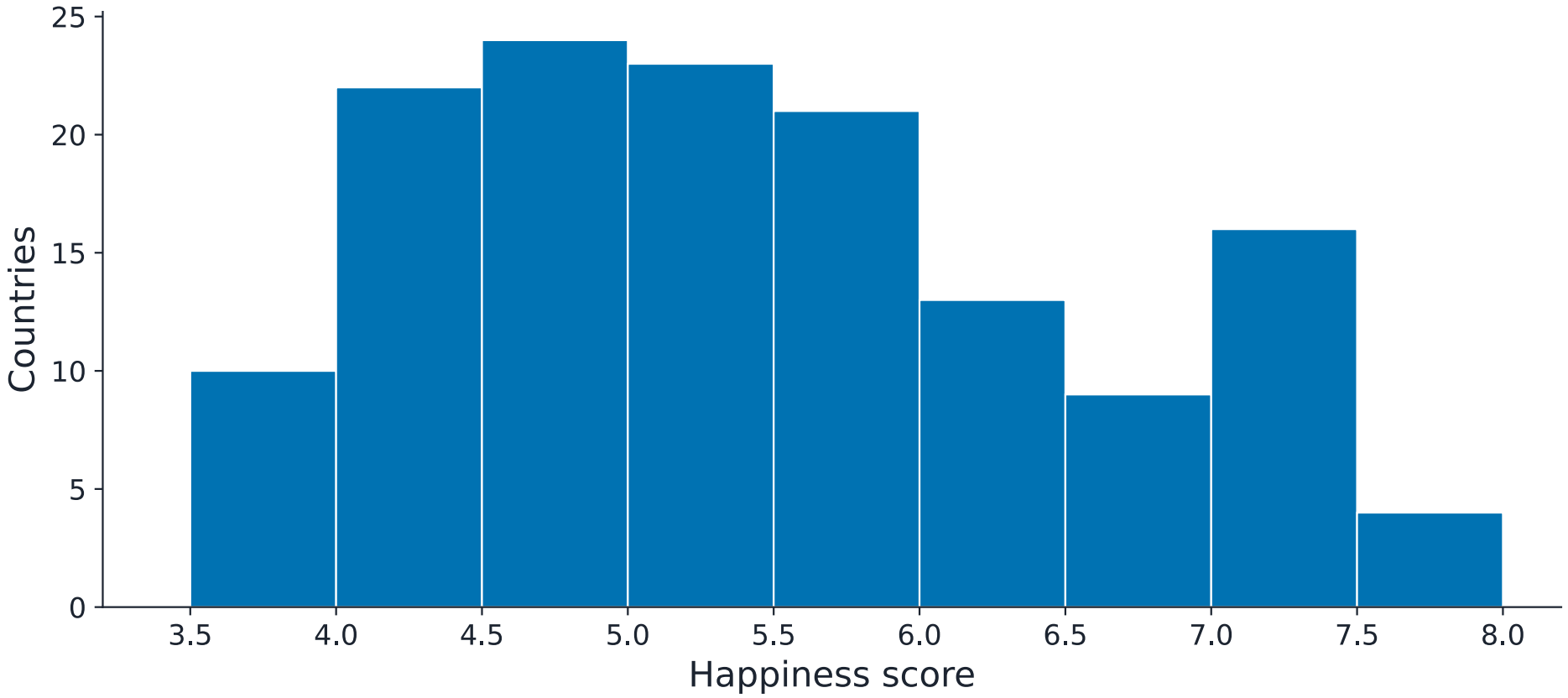
# Is Denmark alone at the top?

Denmark has the highest score, 7.69. Benin has the lowest, 3.53.

EXAMPLE 4 · HISTOGRAM

# Most countries sit between 4 and 6

Ninety of the 142 countries score at least 4 and below 6.




Each bar is half a point wide. The tallest bar, from 4.5 up to 5, holds 24 countries.

EXAMPLE 4 · DENSITY CURVE

# No. Twenty countries score 7 or higher

Denmark is the highest of that group, at 7.69. It is not a country sitting by itself.




The curve peaks near 5 and rises again near 7. Gaussian kernel, bandwidth 0.35. Benin is the lowest score, 3.53.

EXAMPLE 5

# Does money move with the score?

Log GDP per capita against the happiness score.

QUESTION

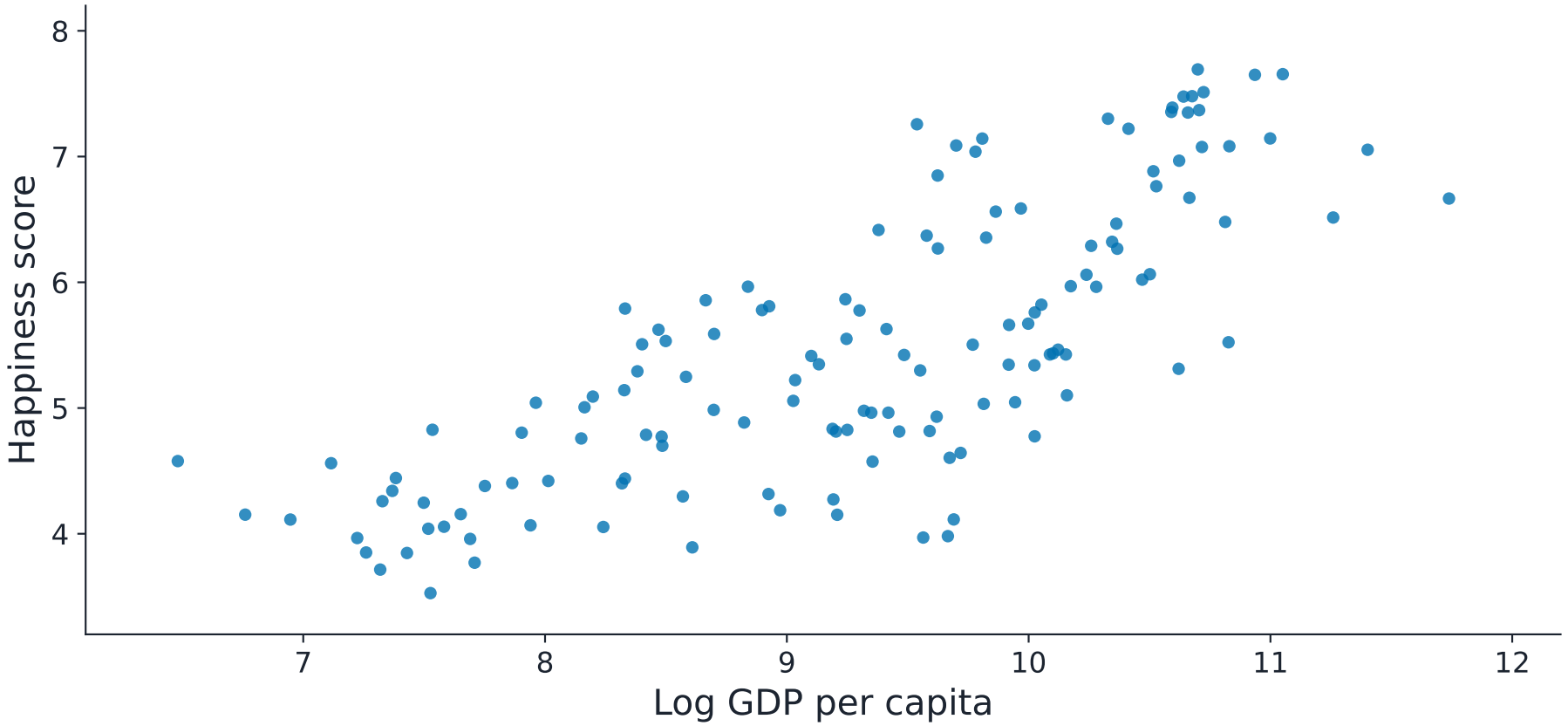
# Does a higher GDP always mean a higher score?

A larger log GDP is a richer country. Three of the 142 countries have no log GDP in this table.

EXAMPLE 5 · SCATTER

# Scores rise with log GDP

The correlation is 0.77 across the 139 countries that have both numbers.

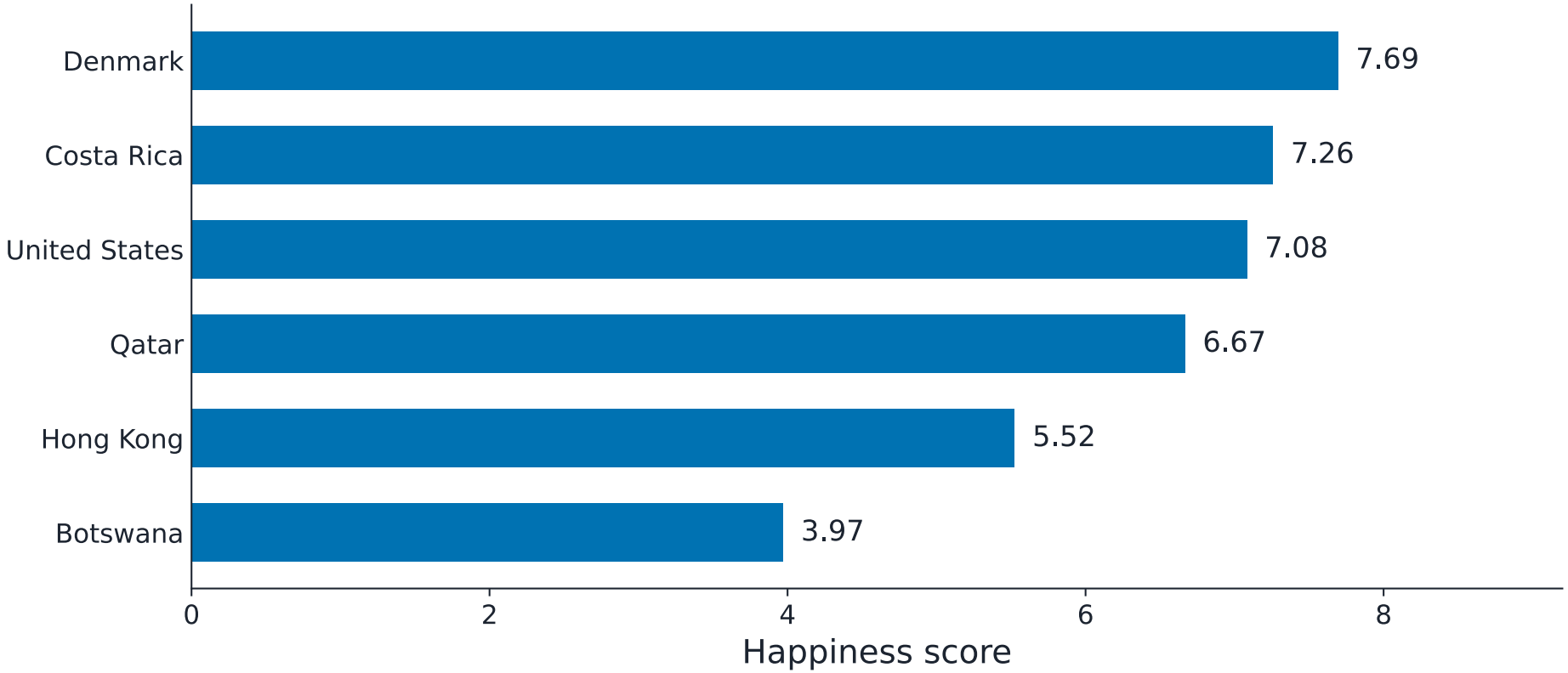



The three countries with no log GDP are left off. A blank is not a zero.

EXAMPLE 5 · BAR

# A higher GDP is not a promise

Qatar has the highest log GDP, 11.7, and a score of 6.67.




Hong Kong and the United States both have log GDP near 10.8. Their scores are 5.52 and 7.08. Costa Rica and Botswana are both near 9.5. Their scores are 7.26 and 3.97.

EXAMPLE 6

# Two halves

The countries are split by log GDP. One half is lower. One half is higher.

QUESTION

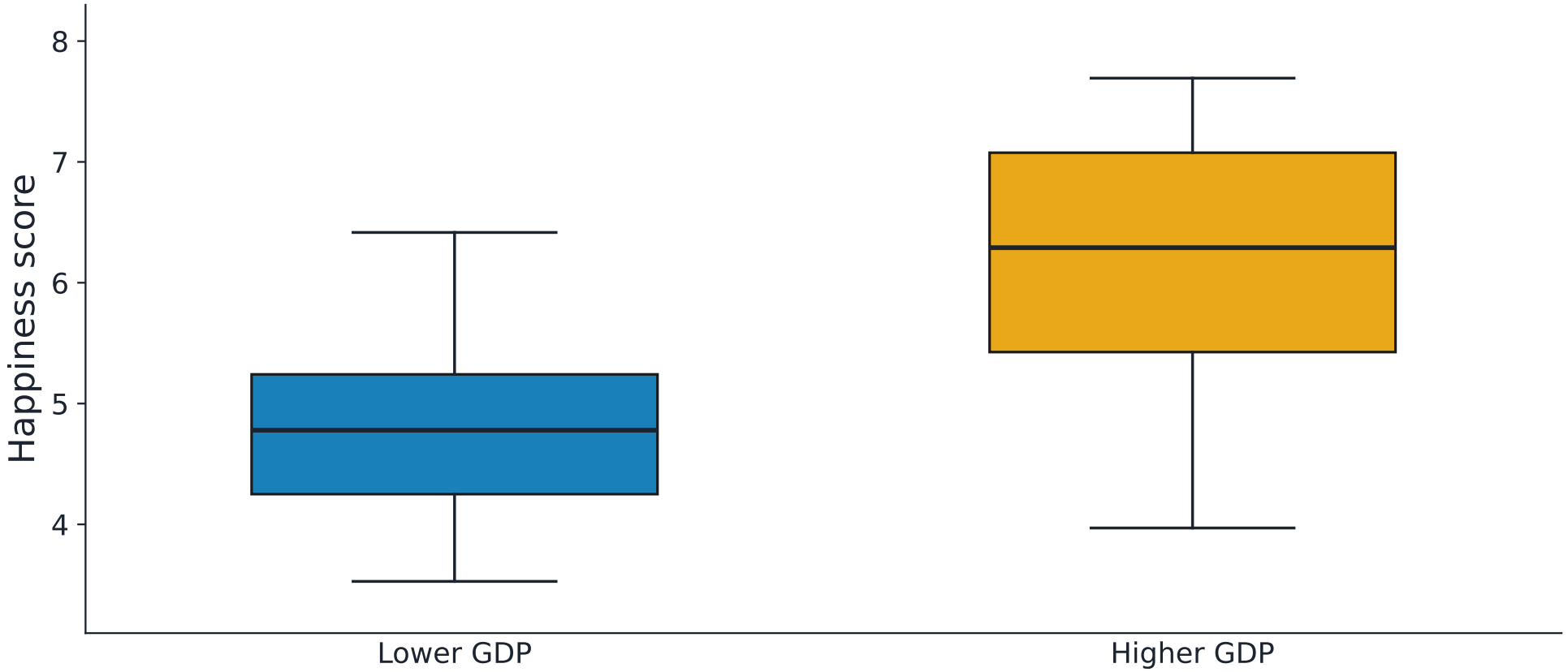
# Which half has the higher happiness scores?

On the next charts, blue is Lower GDP and gold is Higher GDP.

EXAMPLE 6 · BOX PLOT

# Higher GDP has the higher scores

Its box sits above the Lower GDP box.

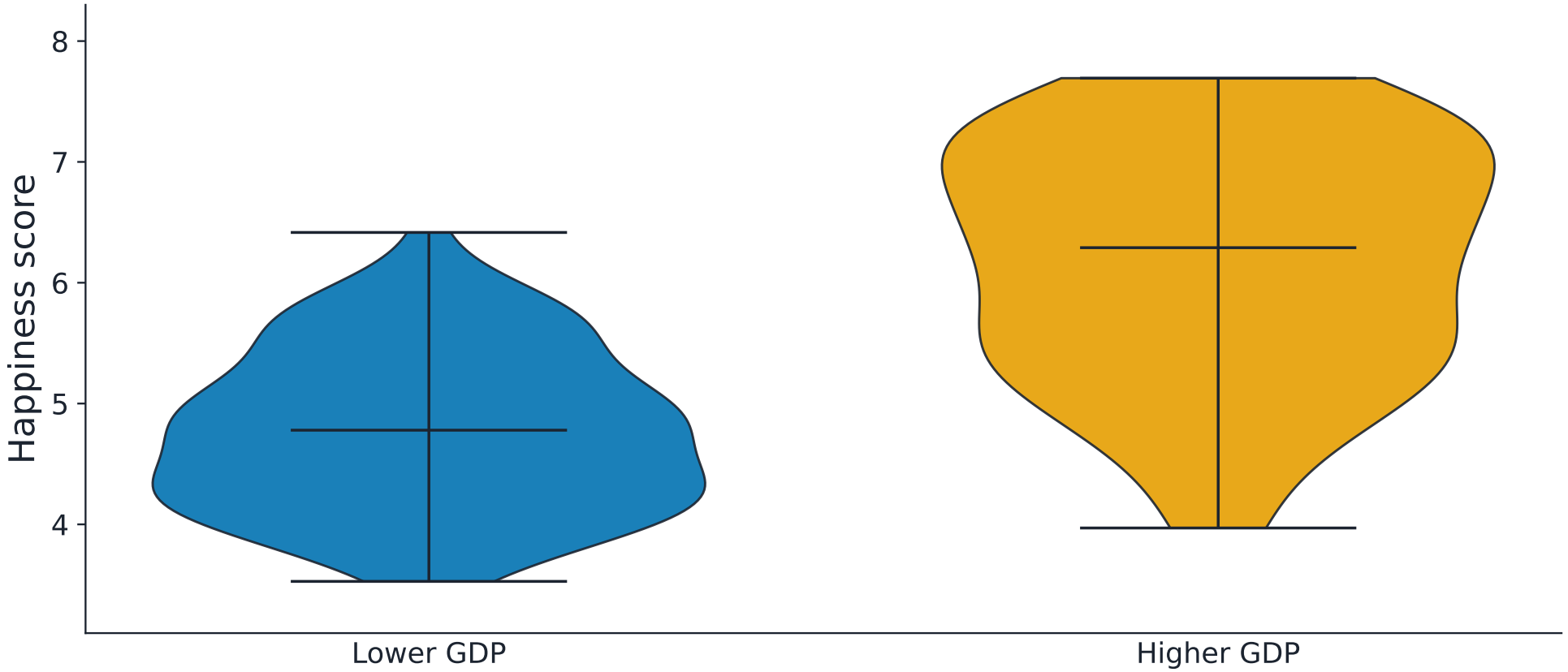



The Higher GDP box runs from 5.4 to 7.1. The Lower GDP box runs from 4.3 to 5.2. Botswana is in the Higher GDP half and scores 3.97.

EXAMPLE 6 · VIOLIN

# The two shapes still overlap

Higher GDP piles up higher. Lower GDP piles up lower.




Same 139 countries. Higher GDP reaches above 7. Lower GDP stops at 6.42, in Colombia.

EXAMPLE 7

# The Broad Street pump

In 1854, cholera deaths in Soho, London, cluster on John Snow’s map. The table marks 578 deaths and 13 public water pumps.

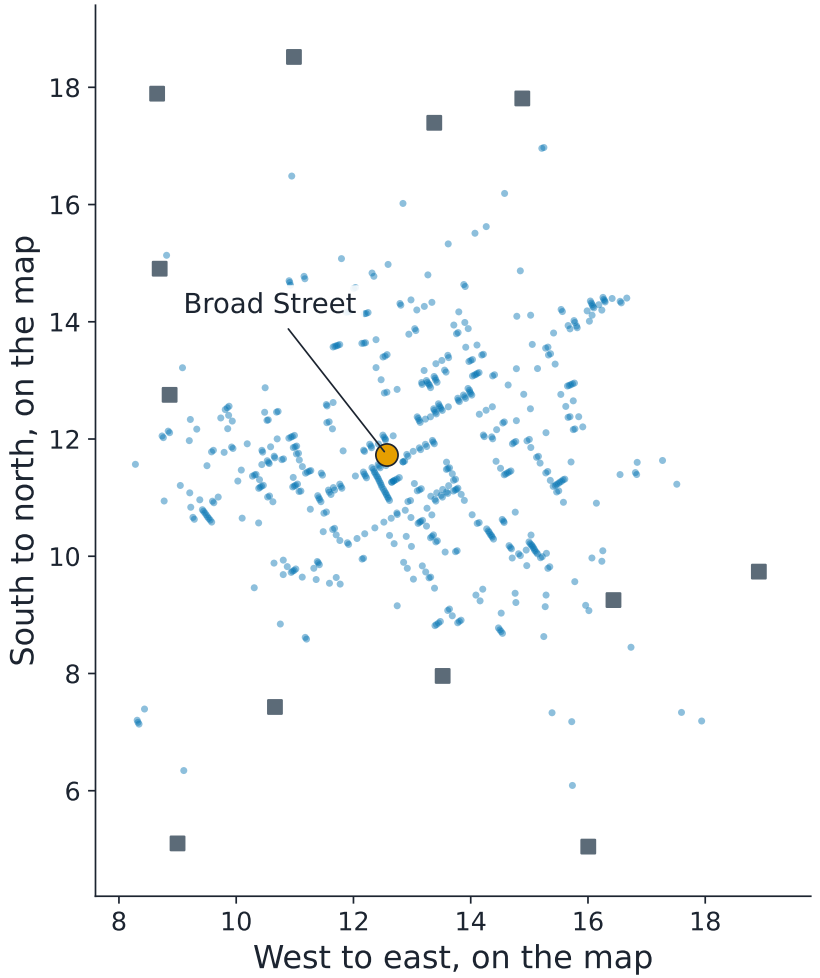
QUESTION

# Which pump are most of these deaths closest to?

The axes are position on Snow’s map. They are not miles.

EXAMPLE 7 · MAP

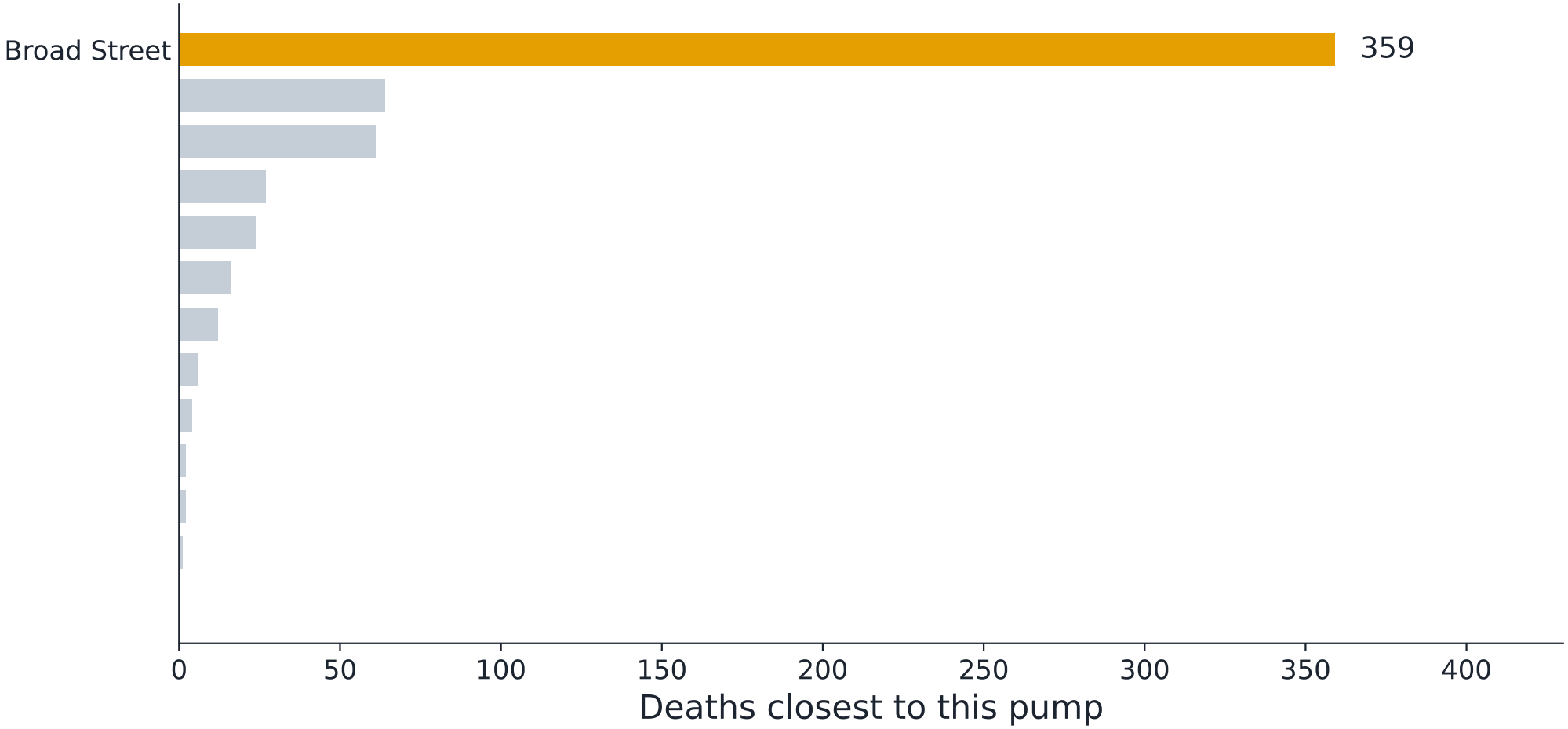
# The deaths cluster around one pump




Each dot is one death. The gold mark is the Broad Street pump. The gray squares are the other 12 pumps.

EXAMPLE 7 · BAR

# 359 of the 578 deaths are closest to Broad Street




The next pump has 64. One pump has none. Closest means a straight line on this map, not the streets a person would walk.

THE END

# Seven results

- Gentoo bills are longer and shallower than Adelie bills. Chinstrap bills are long, and they are not shallow.
- Jupiter is 11.2 Earths wide. Saturn’s rings are not in that chart.
- Death Valley is hotter most of the year. Houston is warmer in January, November, and December.
- Denmark is not alone at the top. Twenty countries score 7 or higher.
- Scores rise with log GDP, and a higher GDP is not a promise.
- The Higher GDP half has the higher scores, and the two shapes still overlap.
- 359 of the 578 deaths are closest to the Broad Street pump.